

Trapezoid Formula Builder Puzzle

Unit 5: Architecture Firm • Lesson 5-2 • EduWonderLab

UNIT 5 • LESSON 5-2 • TEACHER NOTES

Standard 6.G.1

FORMAT Formula-building puzzle	TIME 25–30 min
GROUPING Pairs	TOPIC Area of Trapezoids
STANDARD 6.G.1	PREP LEVEL Medium — print trapezoid cutouts and formula strips

Learning goal *I can find the area of a trapezoid using $A = \frac{1}{2}(b_1 + b_2)h$.*

Teacher setup

- Print one set per pair. Students pair two congruent trapezoids to form a parallelogram.
- They use the puzzle strips to build the area formula, then solve the cards.
- Label b_1 , b_2 , and h clearly on each figure.

Materials

Trapezoid cutouts, Formula puzzle strips, Task cards, Pencil.

Differentiation & language support

- Formula strip: $A = \frac{1}{2}(b_1 + b_2) \times h$ (SPED).
- Sentence stem: “I added the two bases first because ____.”
- ESOL: label the two bases as “top” and “bottom.”

Key vocabulary (English / Español):

Term	Español	What it means
trapezoid	trapecio	a quadrilateral with one pair of parallel sides
bases	bases	the two parallel sides

Quick teacher check: Ask a pair to show the “add the bases first” step on Card 5.

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Name: _____ Date: _____ Class: _____

Your goal: *I can find the area of a trapezoid using $A = \frac{1}{2}(b_1 + b_2)h$.*

What you need

Trapezoid cutouts, Formula strips, Task cards.

How to play

1. Pair two trapezoids to form a parallelogram and see where $\frac{1}{2}$ comes from.
2. Add the two bases ($b_1 + b_2$).
3. Multiply by the height, then take half.
4. Record the area with square units.

Trapezoid Cards

Add the two bases, multiply by the height, then take half. $A = \frac{1}{2}(b_1 + b_2)h$.

1. $b_1 = 6, b_2 = 10, h = 4$	2. $b_1 = 5, b_2 = 9, h = 6$
3. $b_1 = 8, b_2 = 12, h = 5$	4. $b_1 = 3, b_2 = 7, h = 4$
5. $b_1 = 10, b_2 = 14, h = 8$	6. $b_1 = 4, b_2 = 6, h = 3$
7. $b_1 = 7, b_2 = 11, h = 10$	8. $b_1 = 6, b_2 = 8, h = 5$

Record your work ($\frac{1}{2}(b_1 + b_2)h$):

Explain your thinking

Explain why the trapezoid area formula includes “one half.”

Sentence frame: *I added the two bases first because the formula uses ____.*

★ **Challenge:** A trapezoid has area 40 sq units, bases 6 and 10, and unknown height. Find h . ($h = 5$)

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. $b_1 = 6$, $b_2 = 10$, $h = 4$ Answer: 32 sq units	2. $b_1 = 5$, $b_2 = 9$, $h = 6$ Answer: 42 sq units
3. $b_1 = 8$, $b_2 = 12$, $h = 5$ Answer: 50 sq units	4. $b_1 = 3$, $b_2 = 7$, $h = 4$ Answer: 20 sq units
5. $b_1 = 10$, $b_2 = 14$, $h = 8$ Answer: 96 sq units	6. $b_1 = 4$, $b_2 = 6$, $h = 3$ Answer: 15 sq units
7. $b_1 = 7$, $b_2 = 11$, $h = 10$ Answer: 90 sq units	8. $b_1 = 6$, $b_2 = 8$, $h = 5$ Answer: 35 sq units

Teaching notes & look-fors

- Two congruent trapezoids form a parallelogram, which is why we take half.
- Challenge: $40 = \frac{1}{2}(16)(h) \rightarrow 40 = 8h \rightarrow h = 5$.